Tutorial Questions 04 – The derivative Dr Killian O'Brien
6G4Z3006 Calculus 07/04/22

Continuity

1. **Quotient case of theorem 4.3:** Suppose $g(x)$ is non-zero in some interval of the domain containing the point a and that g and another function f are continuous at a. Prove that the quotient function f/g defined by $(f/g)(x) = f(x)/g(x)$ is continuous at $x = a$. You should follow the principles used for the similar result of the quotient case of the algebra of limits theorem for quotients.

Solution:
Since we already know that continuity is preserved by products of functions, to prove that f/g is continuous if f and g are we only have to prove that the reciprocal part $1/g$ is continuous as the quotient f/g can be expressed as the product $f/g = f \times (1/g)$.
So we assume that $g(a) \neq 0$ and that g is continuous at a and that $\epsilon > 0$ is given. As g is continuous at a we can also assume that g is non-zero in some interval centred at a (more on this later). Notice the relations

$$|1/g(x) - 1/g(a)| = \left|\frac{g(a) - g(x)}{g(x)g(a)}\right| = \frac{1}{|g(x)||g(a)|}|g(x) - g(a)|.$$

So to constrain the size of $|1/g(x) - 1/g(a)|$ we need to constrain the size of $|g(x) - g(a)|$ (which we can do with our freedom to choose δ) and prevent the denominator $|g(x)||g(a)|$ from getting too small, i.e. find a lower bound for it (which we can do using the continuity of g).
We know there exists $\delta_1 > 0$ such that if $|x-a| < \delta_1$ then $|g(x)-g(a)| < |g(a)|/2$ which implies

$$|g(x)| > \frac{|g(a)|}{2}.$$

We also know there exists $\delta_2 > 0$ so that is $|x - a| < \delta_2$ then

$$|g(x) - g(a)| < \frac{\epsilon|g(a)|^2}{2}.$$

Now let $\delta = \min(\delta_1, \delta_2)$ so that if $|x - a| < \delta$ we have both the above inequalities and we can say

$$|1/g(x) - 1/g(a)| < \frac{2}{|g(a)||g(a)|}\frac{\epsilon|g(a)|^2}{2} = \epsilon,$$

as required for $1/g$ to be continuous at a.

2. Use the formal definition of continuity to show that the function $f(t) = t^2$ is continuous at every point in its domain.

Solution:
Let $a \in \mathbb{R}$ and let $\epsilon > 0$ be given. We need to constrain the size of $|t^2 - a^2|$ using our ability to constrain the size of $|t - a|$ with our freedom to choose any $\delta > 0$.

Notice the following relations

$$\begin{aligned}|t^2 - a^2| &= |(t-a)(t+a)| \\ &= |t-a||t+a| \\ &= |t-a||t-a+2a| \\ &\leq |t-a|\left(|t-a| + 2|a|\right) \\ &= |t-a|^2 + 2|a||t-a|.\end{aligned}$$

So as long as we choose $\delta < \min(\sqrt{\epsilon/2}, \epsilon/(4a))$, say for instance choose

$$\delta = \frac{1}{3}\min(\sqrt{\epsilon/2}, \epsilon/(4a)),$$

then we will have

$$|t^2 - a^2| < (\sqrt{\epsilon/2})^2 + 2|a|(\epsilon/(4a)) = \epsilon/2 + \epsilon/2 = \epsilon,$$

as required. So we can conclude that $f(t) = t^2$ is continuous at this a and hence everywhere.
Another approach to this would be to use the fact that the function $f(t) = t$ is continuous everywhere (which is much easier to prove) and then use the product form of theorem 4.3 to conclude that $t^2 = t \cdot t$ is also continuous.

The derivative

3. Can you confidently quote the limit definition of the derivative from memory? Can you confidently quote it using different notation than the usual f and x? For instance what is the definition for the derivative of a function λ of a real variable y at a point c of its domain?

Solution:
The definition would be

$$\left.\frac{d\lambda}{dy}\right|_{y=c} = \lim_{y\to c}\frac{\lambda(y) - \lambda(c)}{y - c}.$$

Or in the alternative form,

$$\left.\frac{d\lambda}{dy}\right|_{y=c} = \lim_{h\to 0}\frac{\lambda(c+h) - \lambda(c)}{h}.$$

4. Use the limit definition of the derivative to obtain the derivative of the function defined by $g(t) = t^5$.

Solution:

$$\begin{aligned}g'(a) &= \lim_{t\to a}\frac{t^5 - a^5}{t-a} \\ &= \lim_{t\to a}\frac{(t-a)(t^4 + t^3a + t^2a^2 + ta^3 + a^4)}{t-a} \\ &= \lim_{t\to a}(t^4 + t^3a + t^2a^2 + ta^3 + a^4) \\ &= 5a^4\end{aligned}$$

This holds for each a so the derivative function is $g'(t) = 5t^4$.

5. Generalise the previous question to show that for each $n > 0$ if $h(t) = t^n$ then $h'(t) = nt^{n-1}$.

Solution:

$$\begin{aligned} h'(a) &= \lim_{t \to a} \frac{t^n - a^n}{t - a} \\ &= \lim_{t \to a} \frac{(t-a)\left(\sum_{j=0}^{n-1} a^j t^{n-1-j}\right)}{t - a} \\ &= \lim_{t \to a} \left(\sum_{j=0}^{n-1} a^j t^{n-1-j}\right) \\ &= na^{n-1} \end{aligned}$$

This holds for each a so the derivative function is $h'(t) = nt^{n-1}$.

6. Use the limit definition of the derivative to obtain the derivative of the function defined by $h(t) = t^{-2}$.

Solution:

$$\begin{aligned} h'(a) &= \lim_{t \to a} \frac{1/t^2 - 1/a^2}{t - a} \\ &= \lim_{t \to a} \frac{(a^2 - t^2)/t^2 a^2}{t - a} \\ &= \lim_{t \to a} \frac{-(a+t)}{t^2 a^2} \\ &= \frac{-2a}{a^4} \\ &= \frac{-2}{a^3} \end{aligned}$$

This holds for each a so the derivative function is $h'(t) = -2t^{-3}$.

7. Generalise the previous question to show that for each $n > 0$ if $h(t) = t^{-n}$ then $h'(t) = -nt^{-n-1}$.

Solution:

$$\begin{aligned} h'(a) &= \lim_{t \to a} \frac{1/t^n - 1/a^n}{t - a} \\ &= \lim_{t \to a} \frac{(a^n - t^n)/t^n a^n}{t - a} \\ &= \lim_{t \to a} \frac{-\left(\sum_{j=0}^{n-1} a^j t^{n-1-j}\right)}{t^n a^n} \\ &= \frac{-na^{n-1}}{a^{2n}} \\ &= \frac{-n}{a^{n+1}} \end{aligned}$$

This holds for each a so the derivative function is $h'(t) = -nt^{-n-1}$.

8. What does the product rule look like for products of three or more functions? That is, suppose functions f, g, h are all differentiable at a point $x = a$. Find an expression for

the derivative at a of the product function fgh defined by $(fgh)(x) = f(x)g(x)h(x)$, Your expression will feature the derivatives $f'(a), g'(a), h'(a)$ and the values $f(a), g(a), h(a)$. What about a product of four functions? Or for a general product of n functions, i.e. the function defined by

$$(f_1 f_2 \dots f_n)(x) = f_1(x) f_2(x) \dots f_n(x),$$

or expressed using the Pi product notation

$$\left(\prod_{i=1}^{n} f_i\right)(x) = \prod_{i=1}^{n} f_i(x).$$

Solution:
For three functions we have

$$\begin{aligned}(fgh)'(a) &= ((fg)h)'(a)\\ &= (fg)(a)h'(a) + (fg)'(a)h(a)\\ &= f(a)g(a)h'(a) + \big(f(a)g'(a) + f'(a)g(a)\big)h(a)\\ &= f(a)g(a)h'(a) + f(a)g'(a)h(a) + f'(a)g(a)h(a).\end{aligned}$$

So we get a sum of three terms, in each term we have a product of three factors which are the various combinations of one of the functions differentiated and the rest not, and all aevaluated at a.
This pattern is replicated for products of any number of functions. This result can be proved formally using proof by induction, with the original two-factor product rule as the base case. We can write the result as

$$\left(\prod_{i=1}^{n} f_i\right)'(a) = \sum_{i=1}^{n}\left(f_i'(a)\left(\prod_{\substack{j=1\\ j\neq i}}^{n} f_j(a)\right)\right).$$

9. In a similar spirit to the previous question, what does the chain rule look like for compositions of three or more functions? For instance, what is the derivative of $f \circ g \circ h$ at a point $x = a$, how does it depend on various information about the component functions f, g, h. We can assume that f, g, h are differentiable everywhere.

   Can you generalise to the case of a composition of n functions, where n could be any positive integer?

   Solution:
   For three functions we have

   $$\begin{aligned}(f \circ g \circ h)'(a) &= ((f \circ g) \circ h)'(a)\\ &= (f \circ g)'\big(h(a)\big)h'(a)\\ &= f'(g(h(a)))g'(h(a))h'(a).\end{aligned}$$

10. Use properties of the derivative and results from the table of derivatives to obtain the derivatives of the following functions.

    a) $f(x) = x^5 + 3x^4 - 5x^2 + 10$
    b) $g(x) = 3x^{1/2} - x^{3/2} + 2x^{-1/2}$
    c) $\lambda(t) = \sqrt{3t} + 3\sqrt{t}$
    d) $\mu(y) = (2 + 5y + 1/2y^2)^{1/2}$
    e) $\phi(z) = (z + 1)\sqrt{z^4 - 2z + 2}$
    f) $h(x) = x/\sqrt{1 - 4x^2}$
    g) $p(x) = \sqrt{1 + \sqrt{x}}$

    Solution:
    The derivatives are

    a) $5\,x^4 + 12\,x^3 - 10\,x$
    b) $-\frac{3}{2}\,\sqrt{x} + \frac{3}{2\,\sqrt{x}} - \frac{1}{x^{\frac{3}{2}}}$
    c) $\frac{\sqrt{3}}{2\,\sqrt{t}} + \frac{3}{2\,\sqrt{t}}$
    d) $\frac{y+5}{2\,\sqrt{\frac{1}{2}\,y^2+5\,y+2}}$
    e) $\frac{\left(2\,z^3-1\right)(z+1)}{\sqrt{z^4-2\,z+2}} + \sqrt{z^4 - 2\,z + 2}$
    f) $\frac{4\,x^2}{(-4\,x^2+1)^{\frac{3}{2}}} + \frac{1}{\sqrt{-4\,x^2+1}}$
    g) $\frac{1}{4\,\sqrt{x}\sqrt{\sqrt{x}+1}}$

11. Using geometric arguments it can be shown that

    $$\lim_{h\to 0} \frac{\sin(h)}{h} = 1 \text{ and } \lim_{h\to 0} \frac{1-\cos(h)}{h} = 0.$$

    Use these results and the relation $\sin(A + B) = \cos(A)\sin(B) + \sin(A)\cos(B)$ to carry out the limit definition of the derivative and show that

    $$\frac{d}{dx}\sin(x) = \cos(x) \text{ and } \frac{d}{dx}\cos(x) = -\sin(x).$$

    Many other derivatives of trigonometric functions can be obtained by combining these results with other properties of the derivative.

Solution:

$$
\begin{aligned}
\left.\frac{d}{dx}\sin(x)\right|_{x=a} &= \lim_{h\to 0}\frac{\sin(a+h)-\sin(a)}{h} \\
&= \lim_{h\to 0}\frac{\cos(a)\sin(h)+\sin(a)\cos(h)-\sin(a)}{h} \\
&= \cos(a)\left(\lim_{h\to 0}\frac{\sin(h)}{h}\right)+\sin(a)\lim_{h\to 0}\frac{\cos(h)-1}{h} \\
&= \cos(a), \text{ from given limits.}
\end{aligned}
$$